

SDET Practical Assignment

Software Development Engineer in Test

Role	Software Development Engineer in Test (SDET)
Duration	1 to 2 Working Days
Submission	GitHub Repository + Video Walkthrough (with Audio)

Choose any publicly available REST API (e.g. JSONPlaceholder, PetStore, OpenWeatherMap, GitHub API, PokeAPI). Use it consistently across all tasks. State the API name and base URL in your README. Pick an API that supports full CRUD operations.

1. Task Areas, Stack & Deliverables

# Area	Stack / Tools	Deliverables Required
Web UI Automation (Priority #1 — High)	Selenium WebDriver 4.x · Java · TestNG · Cucumber · Maven · Selenium Grid · Jenkins / GitHub Actions	3–5 Gherkin (.feature) scenarios with step definitions - Page Object Model (POM) design pattern - Maven Surefire + TestNG runner configuration - Screenshot on failure, ExtentReports or TestNG HTML report - README with setup & run instructions
Manual API Testing (Priority #2 — High)	Postman (preferred) / Bruno / SoapUI	- Collection with folders for GET, POST, PUT/PATCH, DELETE endpoints - Pre-request scripts & post-response assertions - Environment variables for base URL, tokens, dynamic values - Exported collection file + result screenshots in repo
API Mock Setup	WireMock / Mockoon / Postman Mock Servers	≥3 stubbed endpoints mirroring chosen API - Show automation tests switched to run against mock - Mock config exported/committed to repo
API Automation Testing (Priority #3 — High)	Rest Assured (Java, preferred) / Karate / Newman CLI	≥10 automated test cases covering all HTTP methods - Validate status codes, body, headers, response time - Data-driven tests (JSON / CSV / Excel) - Allure or ExtentReport published in CI pipeline - GitHub Actions YAML or Jenkinsfile in repo
API Health Monitoring	Datadog / Uptrends / Pingdom (free-tier acceptable)	≥1 health check configured against chosen API - Alert thresholds for response time & availability

# Area	Stack / Tools	Deliverables Required
(Priority #4 — Medium)		Dashboard screenshot or screen-recording included
Load Testing — JMeter (Priority #5 — Medium)	Apache JMeter (primary) · Gatling / k6 (optional bonus)	- JMeter .jmx plan with ≥2 scenarios: baseline (10 users) & stress (50 users) - Thread Groups, HTTP Samplers, Assertions, Listeners - HTML Summary Report committed to repo - Brief observation on response time / error rate / throughput

2. Video Walkthrough (Mandatory)

Duration: 20–40 min · Format: MP4 or Google Drive · Face-cam: optional

Cover in this order:

1. Project structure overview
2. Web UI Automation demo
3. Manual API Testing demo
4. Mock API demo
5. API Automation test run
6. CI/CD pipeline execution
7. JMeter load test & HTML report walkthrough
8. API Monitoring dashboard

4. Submission Checklist

Deliverable	Details
GitHub Repo	All code, configs, feature files, .jmx plan, Postman collection, README
Video Walkthrough	20–40 min screen recording with audio narration (YouTube unlisted / Loom / Drive)
CI/CD Config	.github/workflows/*.yml or Jenkinsfile in root of repo
Load Test Report	JMeter HTML Summary Report folder committed to repo
README.md	API chosen + base URL · Setup & run steps · Video link

Submission Email Subject: SDET Assignment — [Your Full Name]

Include: GitHub repo URL · Video link · API name + base URL · Any known limitations or notes